

BENCHMARK: INACCURATE

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457339.6815 +/- 0.0052 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.166 +/- 0.018                 |
| Transit depth (Rp/Rs)^2                           | 2.75 +/- 0.58 [%]               |
| Orbital Inclination (inc)                         | 88.15 +/- 1.34                  |
| Airmass coefficient 1 (a1)                        | 1.186 +/- 0.024                 |
| Airmass coefficient 2 (a2)                        | -0.118 +/- 0.01                 |
| Scatter in the residuals of the lightcurve fit is | 1.98 %                          |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 10.95                           |
| Optimal Annulus                                   | 26.47                           |
| Transit Duration (day)                            | 0.0894 +/- 0.0048               |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.166 ± 0.018 vs 0.1404 (deviation 1.42σ) |
| Duration fit vs published | 0.0894 d vs 0.0754 d (deviation 2.91σ)    |
| Mid-time O–C              | -29.9 min                                 |
| Depth SNR                 | 9.2                                       |
| Residual scatter          | 1.98 %                                    |

Light curve

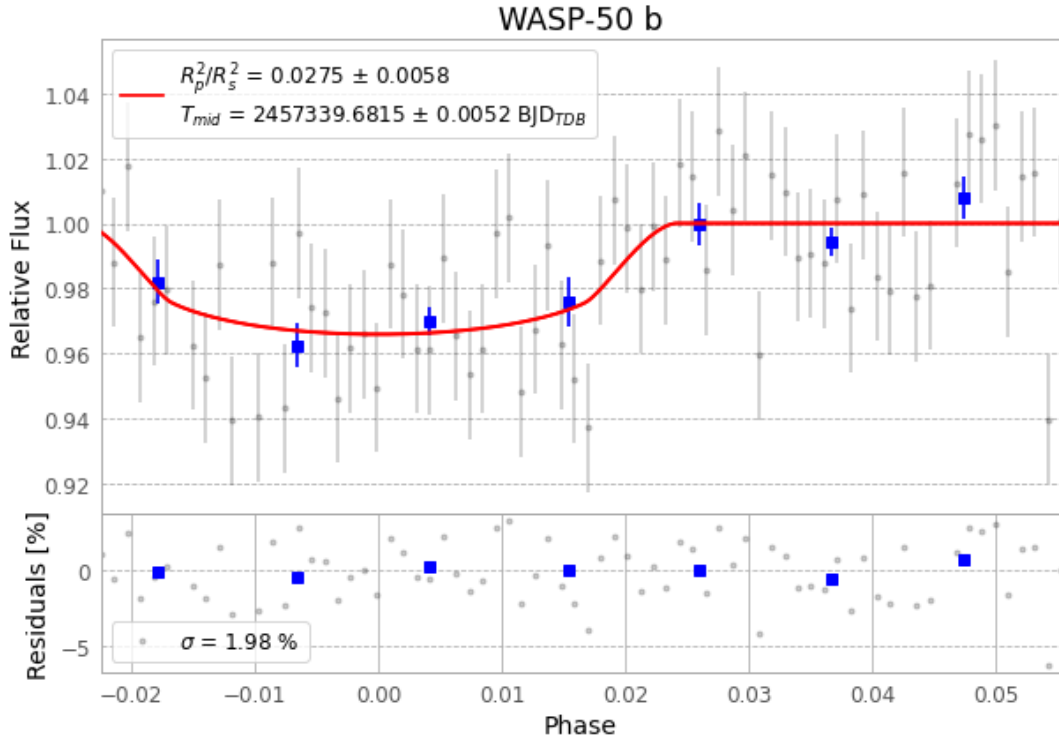


Figure 1 — FinalLightCurve\_WASP-50 b\_2015-11-12.png (SHA-256 92387b9dbc0e3c76...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [134, 329], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 fd2b28c001025ad0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

74 FITS frames (49 MB), WASP-50151113030912.FITS → WASP-50151113064812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-151113.log (e9c55246044e...), FinalLightCurve\_WASP-50 b\_2015-11-12.png (92387b9dbc0e...), FinalLightCurve\_WASP-50 b\_2015-11-12.csv (4f8756eae0be...)

## Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 16:29Z.